

This is a group project.

Data-level parallelism Integrating project:

1. Project proposal detailing the usage of data level parallelism (GPU/CUDA (SIMT) or SIMD-based AVX/AVX2/AVX512 assembly language program). The project can be one of the modules of your planned thesis project which can serve as proof of concept.
 - a. Description of the project
 - b. Proposed algorithm should be able to transform to expose the capability to use SIMT/SIMD parallel computing paradigm.
 - c. References
2. Implementation of the proposed project
3. Presentation pitch of the proposed project

Refer to Canvas for the actual deadline:

Milestone #1: Week 6: Project proposal

Milestone #2: Week 10: Mid-project update demo

- Specify the mid-project milestone in your proposal

Milestone #3: Week 13: Complete project submission and demo

- Final project deliverables should be placed in GitHub containing the following:
 - 1.) Source code or link to your source code
 - 2.) Discussion of parallel algorithms implemented in your program
 - 3.) Execution time comparison between sequential and parallel
 - 4.) Conference paper (optional)

Expanded outline:

- 0.) Abstract of your proposed project
- 1.) Description of your proposed project
 - a. What are the inputs to your project?
 - b. Propose process
 - i. What processes will be parallelized using SIMT (GPU/CUDA) or SIMD-based AVX/AVX2/AVX512 assembly project?
 - ii. What existing implementations have been done with the proposed process (i.e., related works)? How would it differ from your proposed implementation if it has been parallelized?
 - iii. Discuss plans on how to parallelize your proposed process.
 - iv. What other processes are in the project but will not be parallelized
 - c. What are the outputs of your project?
- 2.) Proposed target implementation platform of the proposed project
- 3.) Mid-project milestones of the proposed project
- 4.) References